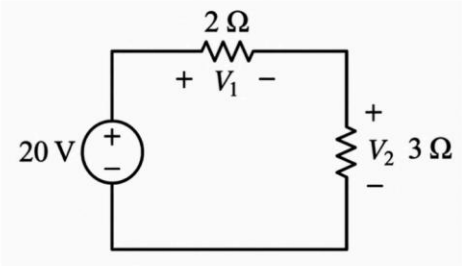
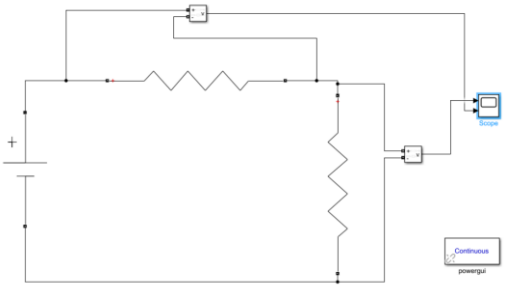
LAB REPORT 09
Simscape Electrical Circuit Simulations

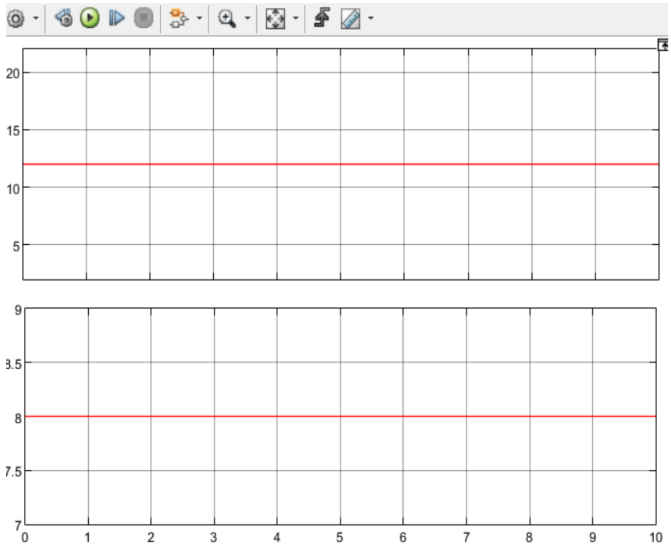
Example 01: Simple DC Circuit Simulation

> Find voltages V1 and V2 in the simple DC circuit.

Example 01 Find voltages V_1 and V_2





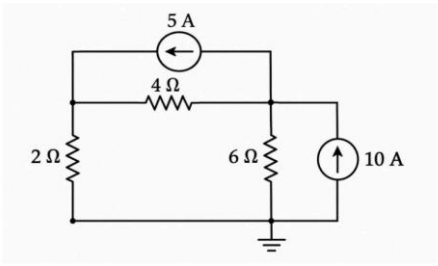


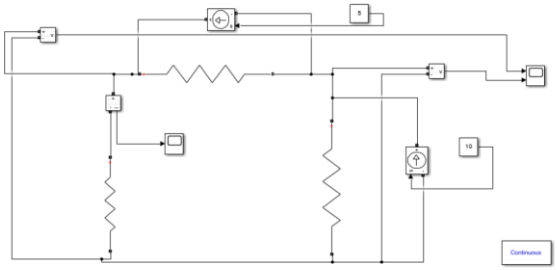
V

Example 02: DC Circuit with Current Sources

> Calculate the node voltages in the given circuit using DC current sources.

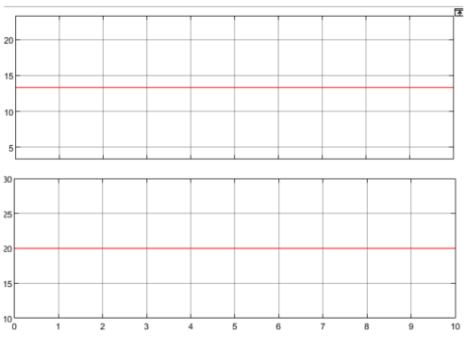
Example 02 Calculate the node voltages in the following circuit:

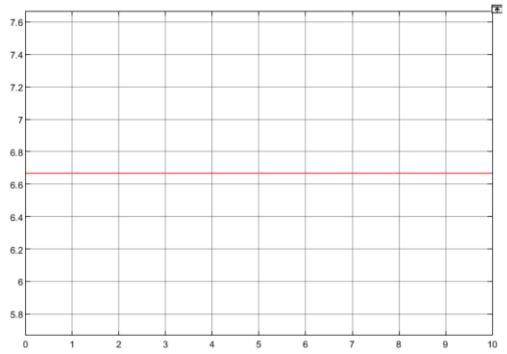




Current : I = 6.666 A

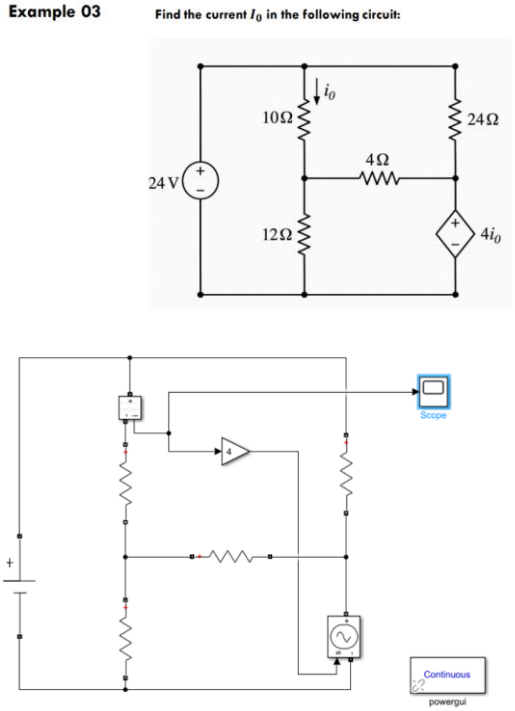
Voltage : V1 = 13.33 v and V2= 20 v



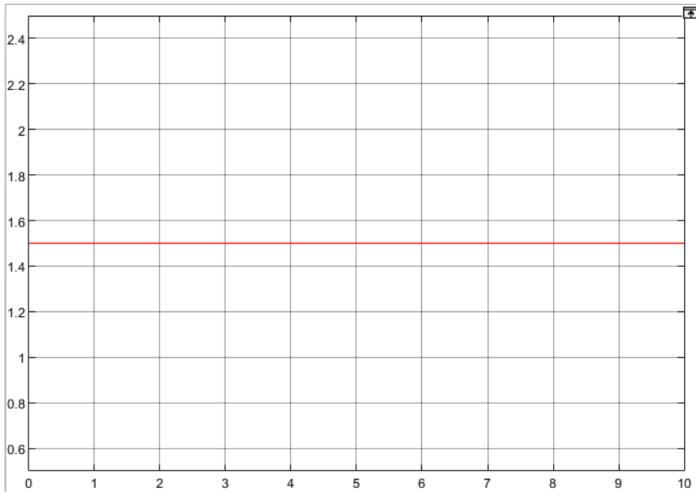


Example 03: Circuit with Dependent Source

Find the current i0 in the circuit featuring a dependent voltage source.



Current : I = 1.5 A



Example 04: Inverting Amplifier (Example 5.3)

Calculate the output voltage vo and the current in the 10-kΩ resistor for the inverting op-amp circuit.

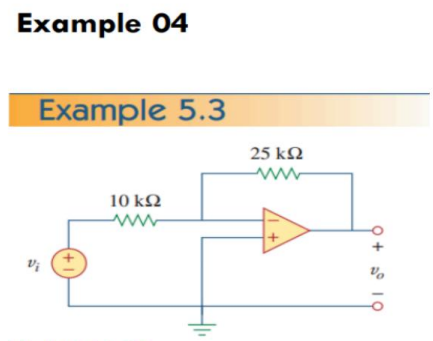
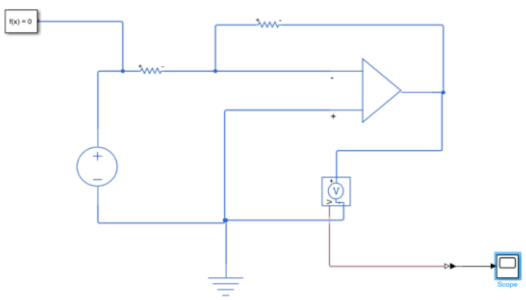
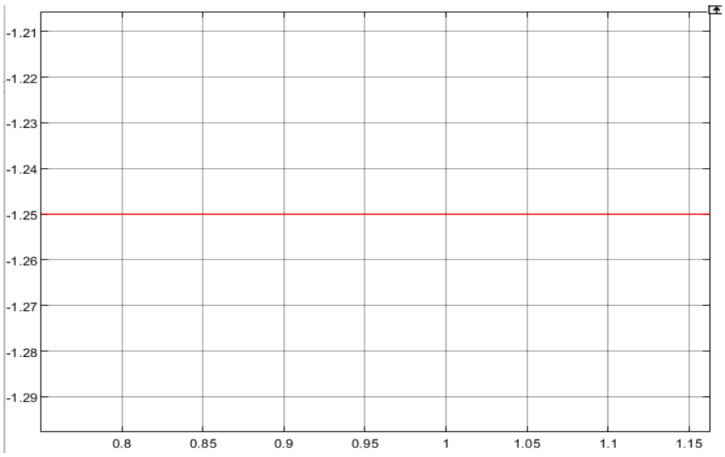


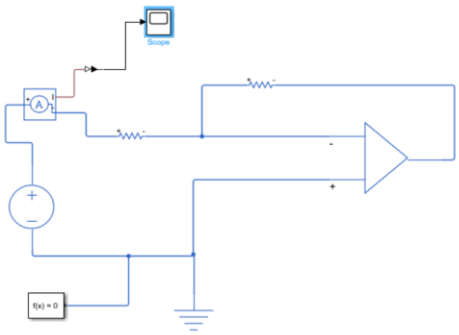
Figure 5.12
For Example 5.3.

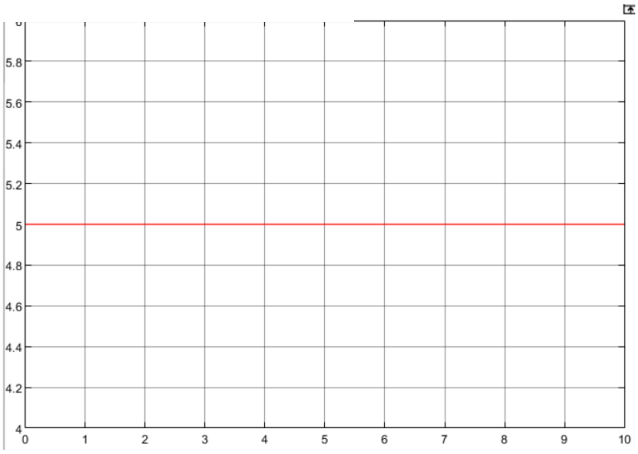


(a) Output Voltage is V = -1.25 V



(b) Current through the 10K ohm is = 50 Micro Ampere





Example 05: Op-Amp Circuit Analysis (Example 5.5)

> Calculate the output voltage vo for the given op-amp circuit with multiple DC sources.

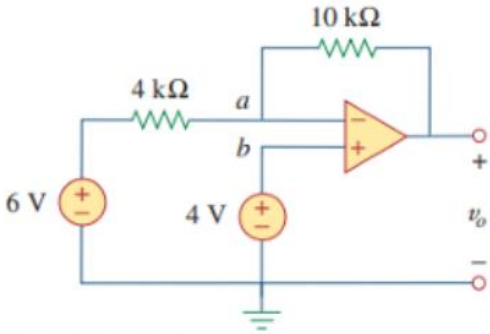
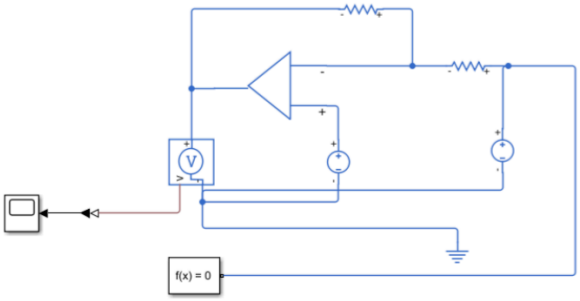
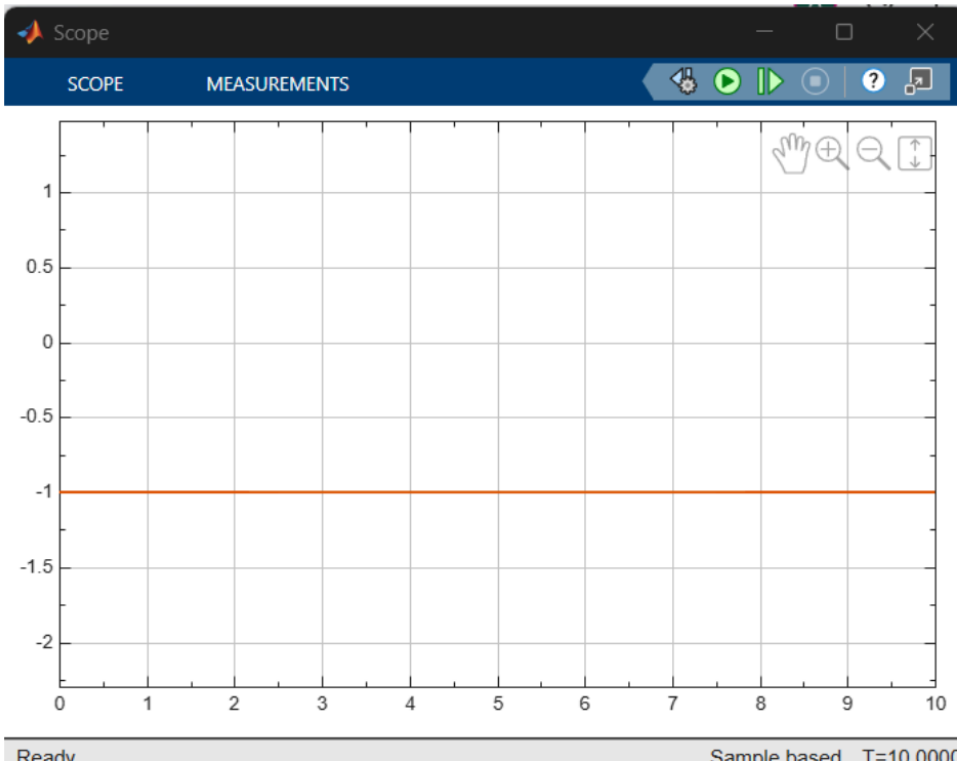


Figure 5.19
For Example 5.5.





Example 06: Advanced Op-Amp Circuit (Example 5.6)

> Calculate vo and io in the given op-amp circuit using Simulink-PS components.

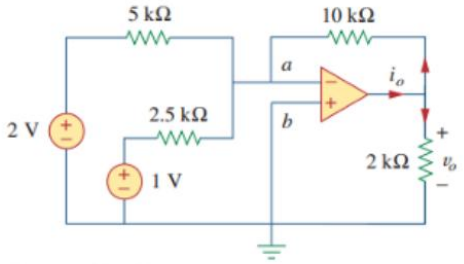
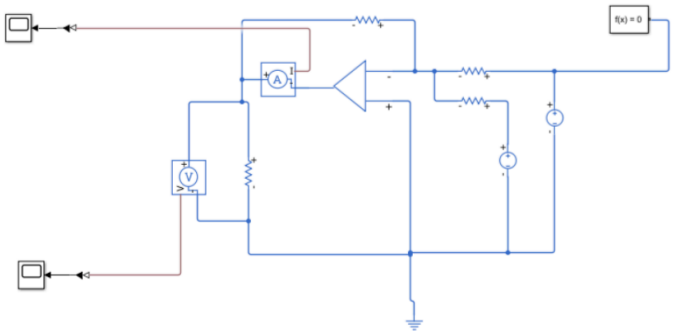
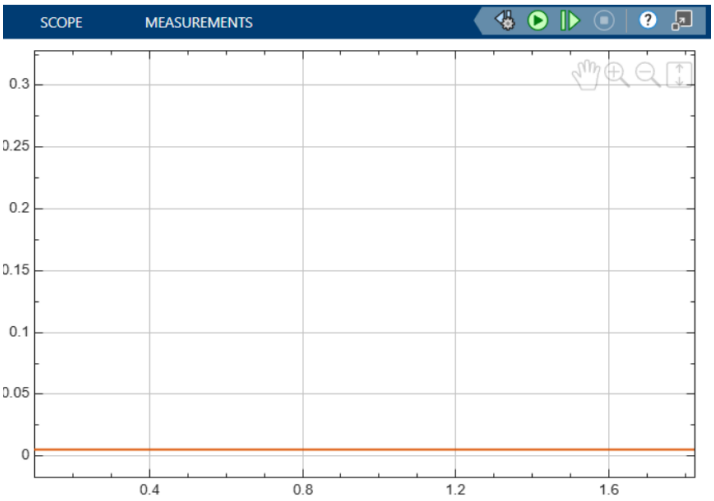
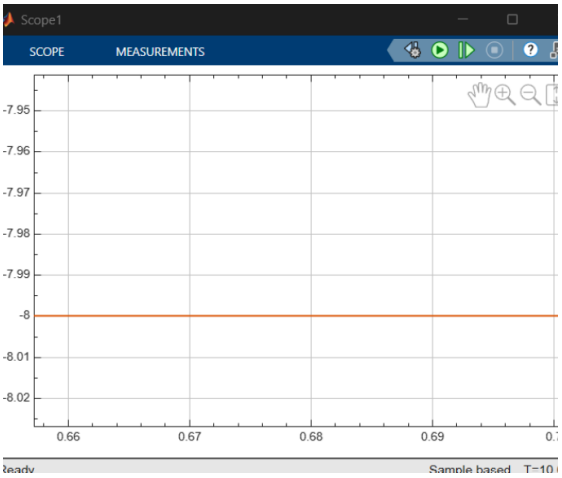


Figure 5.22
For Example 5.6.



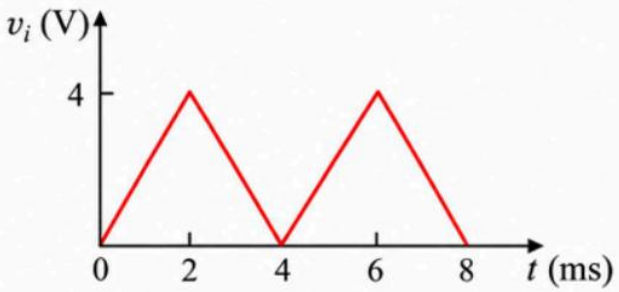


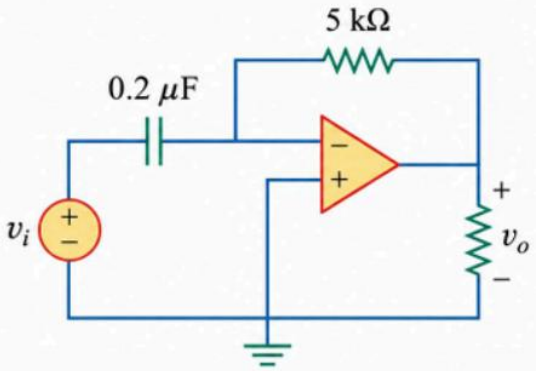


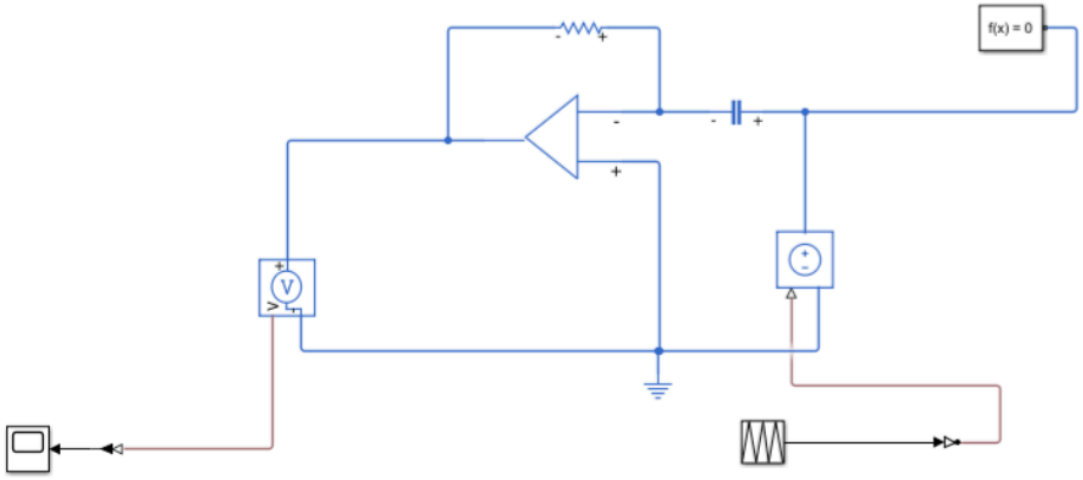
Example 07: Op-Amp Integrator with Capacitor

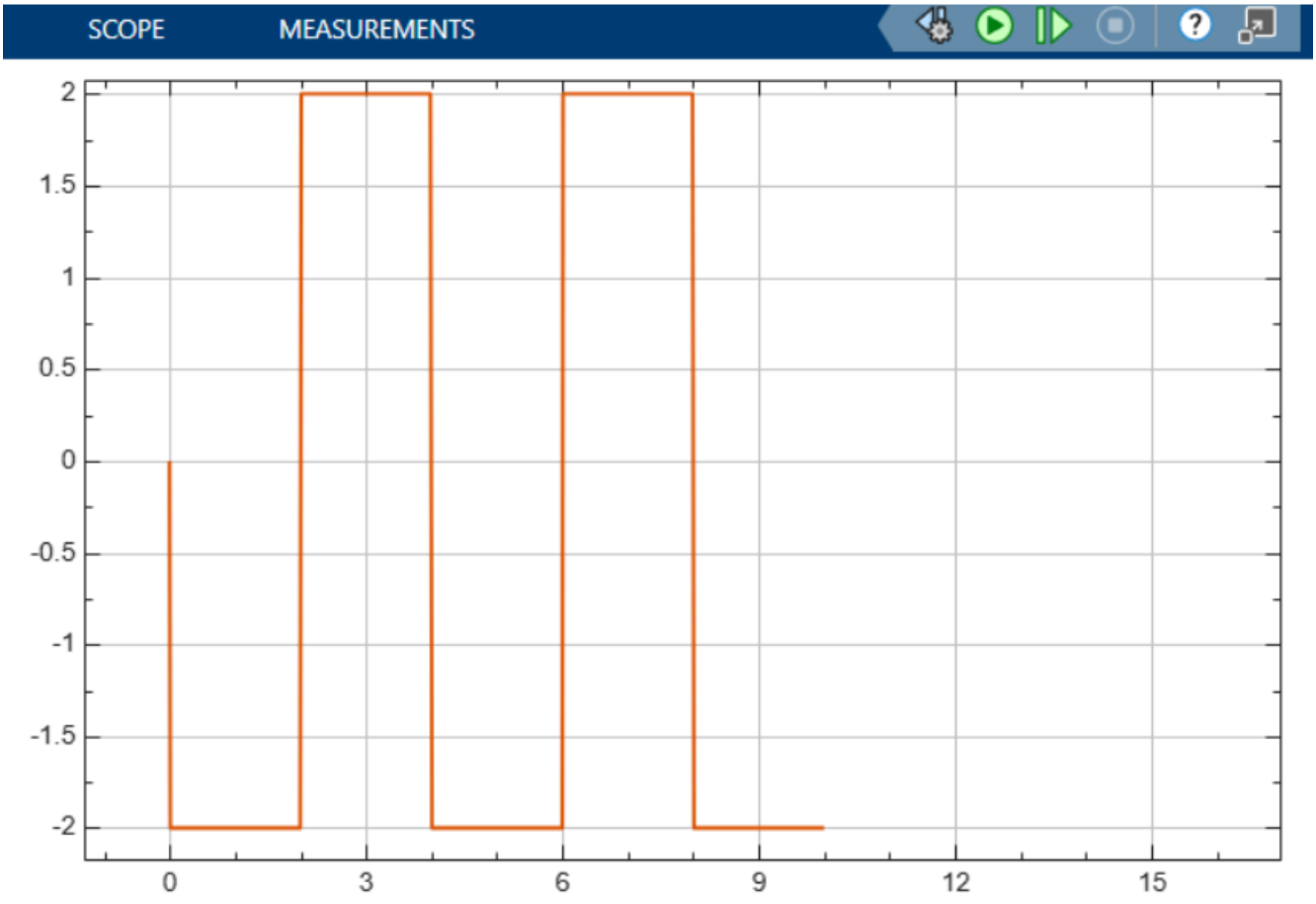
> Sketch the output voltage for the circuit, given the triangular input voltage vi. Take vo = 0 at t = 0.

Sketch the output voltage for the circuit, given the input voltage. Take $v_o = 0$ at $t = 0$.






f(x) = 0


SCOPE
MEASUREMENTS
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0
3
6
9
12
15